Two Quotients of One Machine

Myhill–Nerode cut complexity and locality cohomology of binary addition

Driven by Dean A. Kulik

September 2026

Abstract

An n-bit modular adder is read two ways, and both readings are computed exactly.

Theorem A (the reader cut). Present the input positions of (a,b) ↦ a+b mod 2^n in an arbitrary order. The minimal Myhill–Nerode state count after a set S of positions has been read depends only on S — never on the order used to reach it — and factorises over the maximal runs of S:

```
                F(S) = PROD over maximal runs [l,r] of S  of  f(l,r)

f(l,r) =  1   if [l,r] = [0,n-1]        L = r - l + 1
          2   if l = 0 and r < n-1
        L+1   if l > 0 and r = n-1
       2L+1   if 0 < l <= r < n-1
```

Memory is a cut functional, not a history functional.

Theorem B (the relation residual). Model the same machine as a Cayley complex on $\mathbb{Z}/2^n$ with generators $2^0,\ldots,2^{n-1}$, declaring a square whenever its two moves touch disjoint bits. Then

```
b_1^disjoint(n) = C(n,2) + (n-1) = (n-1)(n+2)/2
```

Both hold for all n. They are different invariants — one a rank, additive over pairs and identical for every reader; the other a state count, multiplicative over the runs of a cut and different for every reading order — and a once-plausible conjecture equating them is refuted structurally rather than numerically.

What the two share is a sufficient statistic. On the object side the class of an undeclared square keeps only a seam bit and a carry-run depth; on the reader side a run keeps only a first-nontransparent depth and a carry mode. The exponentially large coordinate label — 2^n basepoints there, 4^|S| assignments here — is in both cases the reader's own coordinates and not the object's structure.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

Consequence. The carry recurrence is intrinsic to binary addition. That *one bit* of retained state suffices is intrinsic to the bottom-anchored cut: the lower carry port is pinned at $c_0 = 0$, so nothing is owed and only the outgoing carry survives. Anchor the cut anywhere else and the retained state is not a collection of bits at all, because 2L+1 is odd.

1. Introduction

A serial adder carries one bit. The question this paper answers is where that bit comes from — whether it is a fact about addition, about the circuit, or about the way the circuit is read.

The investigation began from a different question. A programme on interfaces asks, for a system with declared local relations, how much structure a reader fails to absorb: the residual of a relation predicate, measured cohomologically. A companion question asks how much state a reader must retain to continue deterministically: a Myhill–Nerode quotient. Both are quotients of one transition system, and it was natural to conjecture that they measure the same thing — that the minimal retained memory counts the relation classes the predicate leaves open, or their orbits under the reader's own step.

That conjecture is false, and its failure is the organising result. For the n-bit adder the residual is C(n,2) + (n−1), the same number for every reader. The memory depends entirely on the reading order and ranges from one bit to about 0.79n bits on the same arena with the same undeclared relations. The two cannot be equal under any counting map: one is a rank, additive over pair directions; the other is a state count, multiplicative over the runs of a cut. A sharper obstruction settles it — an interior run contributes 2L+1, which is odd, so the retained state of most cuts has no bit-decomposition and there is no register to count.

What survives is better than the conjecture. Both quantities admit exact closed forms for all n, and both proofs turn on the same geometric object: the seam between a resolved carry chain and an unresolved one. The paper states and proves both, reports the verification, and logs the corrections — several of which were caught only because two independent readings of the same quantity disagreed.

2. The machine and its two quotients

2.1 The arena and the locality predicate

Conventions. $G_n = \mathbb{Z}/2^n\mathbb{Z}$ with generating set $\{2^0,\dots,2^{n-1}\}$. For $i < n-1$ the generator gives an edge $x \to x+2^i$. For $i = n-1$, $x+2^{n-1} = x-2^{n-1}$, so that undirected edge is self-inverse and the graph is taken simple. Chains are over $\mathbb{Q}$, computed over a prime field as a certificate; $b_1(K) = \dim_{\mathbb{Q}} H_1$. One orientation is chosen per undirected edge, with $e_i(x)$ the oriented edge $x \to x+2^i$, so that $e_{n-1}(y+2^{n-1}) = -e_{n-1}(y)$. That convention is what the top-case proof of §5.5 uses, so it is fixed here rather than introduced there. Throughout, ∂ is the chain boundary — never a cochain coboundary — and $S_{ij}(x) := \partial r_{ij}(x)$.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

The vertex and edge counts are $|V| = 2^n$ and $|E| = (n-1)2^n + 2^{n-1}$, the second because each of the n−1 lower generators contributes 2^n distinct edges while the top generator's edges are shared in pairs. Hence

```
b_1(graph) = |E| - |V| + 1 = (n-2)*2^n + 2^(n-1) + 1
```

Three predicates declare 2-cells. For a pair i < j and a vertex x, the square r_ij(x) has vertices x, $x+2^i$, $x+2^j$, $x+2^i+2^j$.

- disjoint — declared exactly when the two moves touch disjoint bits at x. Three formulations of that condition agree, and the notebook confirms it on every triple to n = 9: (i) $(x \oplus (x+2^i)) \wedge (x \oplus (x+2^j)) = 0$; (ii) some q in {i,…,j−1} has $x_q = 0$; (iii) bits i..j−1 of x are not all 1. Note what this is not: the generators 2^i and 2^j always occupy distinct bits, so this is not static disjointness of supports but state-dependent carry overlap — the $+2^i$ chain must not reach bit j.
- squares — every commuting square, which is all of them, since addition is abelian.
- squares+carry — additionally the triangles $A_i^2 = A_{i+1}$ for i = 0,…,n−2.

Note what the locality predicate is refusing. Every pair of additions commutes globally. What it declines to certify is not commutation but locality: commutations that hold only through an overlapping carry chain. This is a state-dependent analogue of a Mazurkiewicz independence relation rather than that relation itself: in trace theory independence is a fixed, symmetric, irreflexive relation on an alphabet of actions, whereas here whether A_i and A_j count as independent depends on the basepoint. The related fact that, for token systems, the analogous complex computes the hard-core configuration space is an external theorem (Abrams), cited as context; nothing below proves it.

Lemma (the descent identity). Let Z_1 be the cycle space of the shared 1-skeleton and put $B_D = \text{span}\{\partial\sigma : \sigma \text{ declared}\}$, $B_A = \text{span}\{\partial\sigma : \sigma \text{ any square}\}$, so $B_D \subseteq B_A \subseteq Z_1$. Both complexes have the same 1-skeleton, so $b_1(D) - b_1(A) = \dim B_A - \dim B_D$, that is

```
b_1(D) = b_1(A) + dim(B_A / B_D)
```

and the whole object-side question is the single number dim(B_A/B_D).

2.2 The reader and the cut

The reader receives the input positions of (a,b) one at a time in a fixed order $\pi = (p_1,\ldots,p_n)$; after t steps it knows (a_p, b_p) for p in $S_t = \{p_1,\ldots,p_t\}$. The outputs are the n sum bits of a+b mod 2^n, and a sum bit is emitted as soon as no remaining input can change it. Two partial reads are the same state exactly when every completion gives identical not-yet-emitted outputs. That is the Myhill–Nerode quotient for this transducer; F(S) is the number of its classes.

Two engines compute these. A brute-force enumerator that never mentions carries produces the Myhill–Nerode classes directly; a homology engine that never mentions readers produces b_1. They share no code.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

3. Theorem A — the reader cut law

3.1 The local transfer machine

For a known position q write p_q = a_q ⊕ b_q and g_q = a_q b_q. The carry recurrence is c_{q+1} = g_q ∨ (p_q c_q) and the output is s_q = p_q ⊕ c_q. So every known position is exactly one of three transfer types:

```
(0,0)  kill         c_{q+1} = 0
(1,1)  set          c_{q+1} = 1
(0,1),(1,0) transparent   c_{q+1} = c_q
```

That is the whole local machine. Scanning a maximal run upward from its lower port, the unknown incoming carry survives exactly as long as positions are transparent. At the first nontransparent position the dependence on the incoming carry terminates: every output above it is already determined and need not be retained. Every run therefore has a canonical stopping depth d.

3.2 The four run factors

Let Φ send an assignment on S to its tuple of per-run signatures.

run	what survives	count
interior, 0 < l ≤ r < n−1	depth d in 1..L x carry out in {0,1}, plus all-transparent	2L+1
bottom-anchored, l = 0 < r < n−1	c_0 = 0 known, all outputs emitted; only c_{r+1}	2
top, l > 0, r = n−1	unknown carry in, no observable carry out; only d	L+1
full, [0,n−1]	nothing unresolved	1

The interior count is the product of an owed-output ledger and a carry mode: L owed-prefix lengths times two carry-out values, plus one transparent state that transmits the unresolved carry and owes the whole run. The top run differs from the interior only in that the kill/set distinction after the decisive position has nowhere observable to go, because addition is mod 2^n.

3.3 Gap independence

Between two maximal runs lies at least one unread position, and a single unread position can kill, set or transmit the carry. Assigning the first unread position of a gap to (0,0) or (1,1) forces the carry entering the next run to any chosen value, whatever the run below did. Gaps therefore decouple adjacent run ports; distinct runs have disjoint read positions, so their signatures are chosen independently. This is the step that makes the count multiplicative rather than merely bounded.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

3.4 Both Myhill–Nerode directions

Equal summaries ⇒ equal futures. Process runs upward. Each gap has the same future input in both assignments and the run below has the same carry-transfer signature, so each run sees the same incoming carry; its signature then fixes its unresolved outputs and its outgoing carry. Induction gives identical future output under every completion.

Distinct summaries ⇒ distinguishable. Take the lowest run R = [l,r] whose signature differs, and complete by one recipe: set every unread position transparent (0,1), then, if l > 0, set position l−1 — unread, because R is maximal — to (0,0) or (1,1) to force the incoming carry. The cases are:

- interior run. Both an owed-output difference and an outgoing-carry difference can occur. The lower gap forces whichever incoming carry separates the owed prefixes; a transparent position above the run turns an outgoing-carry difference into an immediate output difference, since its output is $1 \oplus c$.
- top run. There is no observable carry out mod 2^n, so signatures differ only in the stopping depth d, which the lower gap exposes.
- bottom-anchored run. The incoming carry is pinned at $c_0 = 0$ and nothing is owed, so signatures differ only in the outgoing carry, which the transparent position above exposes.
- full run. One class, so the case cannot arise.

At most two completions are needed (c = 0 and c = 1), and both sub-cases fall out of the same family because the default fill is already transparent. Splitting the cases out is a reviewer's correction: the one-sentence version was true but left an external reader to reconstruct which runs have a lower gap and which have an observable carry out.

Hence x ≡_MN y iff Φ(x) = Φ(y); every signature tuple is realizable; and the state count is the product of the per-run counts. ■

3.5 Verification

The equivalence was tested in both directions against the brute-force quotient: no two assignments with equal Φ have different futures, and no two with equal futures have different Φ. Every read set at n = 3,4,5,6 (8, 16, 32 and 64 sets) and 40 sampled sets at each of n = 7,8. Zero mismatches in either direction, and the class count equals the product in every case. Realizability was checked separately on all 32, 64 and 128 read sets at n = 5,6,7.

The recipe above was run as a constructive procedure on every pair of distinct signatures at n = 4,5,6 — 86, 453 and 2280 pairs — and separated all of them, with the lowest differing run falling in the interior case 34, 239 and 1390 times, bottom-anchored 29, 138 and 653 times, and top 23, 76 and 237 times. No pair required a completion outside the recipe.

The closed form was also checked against brute force order by order: all 2, 6, 24 and 120 reading orders at n = 2,3,4,5, zero mismatches, and the named profiles

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

```
n=7  low->high   1 2 2 2 2 2 2 1      peak 2
n=7  high->low   1 2 3 4 5 6 7 1      peak 7
n=7  even/odd    1 2 6 18 36 12 4 1   peak 36
n=7  middle-out  1 3 5 7 9 11 7 1     peak 11
```

4. Corollaries

4.1 Memory is a cut functional

The decomposition mentions only the maximal runs of S. The order in which positions entered S never appears, so F depends on the cut and not on the history that produced it. The sufficient statistic for the reader's state is the geometry of the read/unread boundary.

4.2 LSB-first is the unique minimum for n ≥ 3

The available run factors are 1 (full run only), 2, L+1 and 2L+1. So F(S) = 2 forces exactly one run, and it is either a bottom-anchored interval [0,r] with r < n−1, or the singleton {n−1} with L+1 = 2. There are exactly n such sets, confirmed for n = 2,…,12.

A reading order with peak 2 must pass through such a set at every step 1 ≤ t ≤ n−1. At t = 1 that is {0} or {n−1}. For n ≥ 3 the second branch is a dead end: no F = 2 set of size two contains n−1, since [0,r] with r < n−1 never does. So $S_1 = \{0\}$, the sets are forced to be nested intervals [0,t−1], and the order is 0,1,…,n−1 and nothing else. At n = 2 both orders are optimal. ■

This is a proof rather than an enumeration artifact, and it explains the ripple direction: LSB-first is the only order that keeps the carry interface anchored to one boundary instead of repeatedly opening interior cuts. Exhaustive search agrees — exactly 1 of n! orders attains peak 2 at n = 3,…,8, and 2 of 2 at n = 2.

4.3 The worst order — stated separately and not proved

```
max over orders of peak F  =  4 * 3^((n-3)/2)    n odd,  n >= 3
                              20 * 3^((n-6)/2)    n even, n >= 6
                          (n = 2 -> 2,  n = 4 -> 6)
```

so reading order moves addition from one bit to about 0.79n bits. This is a combinatorial optimisation over run patterns and is not part of the factorisation theorem. It is verified by dynamic programming to n = 60 and is not proved. The exchange sketch: an interior run of length L buys 2L+1 for L+1 positions (the run plus its separating gap), so the density (2L+1)^(1/(L+1)) is maximised at L = 1, giving factor 3 per 2 positions. Odd n tiles exactly with singleton runs at 0,2,…,n−1. Even n cannot, and the cheapest place to absorb the leftover position is lengthening one interior run to L = 2 (×5/3) rather than the top run (×3/2), the bottom run (×1) or a gap (×1).

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

5. Theorem B — the relation residual

5.1 The three predicates and the descent

The register is the calibration baseline: when the moves are bit flips rather than additions, locality and commutation coincide and $b_1 = 0$ under the disjoint predicate. For the adder they part company, and the residual descends across the three predicates:

```
C(n,2) + (n-1)   ->   n-1   ->   0
```

A locality reader certifies neither the overlapping-carry commutations nor the carry identities; declaring all commutations leaves only the carry identities; declaring those closes H^1. The middle term is classical: declaring every commuting square plus the double edge merging $+2^{n-1}$ with -2^{n-1} makes the covering group abelian on generators $2^0,\ldots,2^{n-1}$ with the top generator of order two, and the loop classes are the kernel of the map onto $\mathbb{Z}/2^n$, of rank n−1.

Measured values, from an engine rebuilt from the specification. The 'none' column is b_1 of the bare Cayley graph, $|E| - |V| + 1 = (n-2)2^n + 2^{n-1} + 1$; it is not used by either theorem and is included only to show what the predicates are reducing.

n	none	disjoint	(n−1)(n+2)/2	squares	+carry
2	3	2	2	1	0
3	13	5	5	2	0
4	41	9	9	3	0
5	113	14	14	4	0
6	289	20	20	5	0
7	705	27	27	6	0
8	1665	35	35	7	0
9	–	44	44	–	–
10	–	54	54	–	–

5.2 The lower bound: witness cocycles

For each pair i < j define a 1-cochain supported on generator-i edges only:

```
phi_ij( x -> x + 2^i )  =  [ bits i..j-1 of x are all 1 ]
```

the indicator that the $+2^i$ carry chain at x reaches bit j — precisely the configurations the locality predicate refuses to declare.

Claim A. ϕ_ij vanishes on every declared square. *Proof.* Take the square for (k,l), k < l, at a vertex x where the moves have disjoint support, so the $+2^k$ carry does not reach bit l. Its boundary is e_k(x) +

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

e_l(x+2^k) − e_l(x) − e_k(x+2^l). If i is neither k nor l, every term lies on another generator and all are zero. If i = k, the value is [bits i..j−1 of x all 1] − [bits i..j−1 of x+2^l all 1]; when l ≥ j, adding 2^l leaves bits i..j−1 alone and the terms cancel, and when i < l < j, disjointness says bits i..l−1 of x are not all 1, which adding 2^l cannot change, so both indicators are zero. If i = l, disjointness says the +2^k carry never reaches bit l = i, so bits ≥ i are untouched and the terms cancel. ■

Claim B. dim(B_A/B_D) ≥ C(n,2). *Proof.* By Claim A each ϕ_ij annihilates B_D, so it descends to a linear functional on Z_1/B_D and hence on B_A/B_D. (Stating it this way avoids a category mismatch a reviewer flagged: the ϕ_ij are cochains while B_A/B_D is a quotient of chain spans, and what is being shown is the independence of chain classes, via a pairing.) Form the matrix of the C(n,2) functionals ϕ_ij against the C(n,2) chosen classes [S_kl(x*)], x* = 2^l − 2^k. If it has rank C(n,2) then those classes are independent in B_A/B_D. Its entries are exactly

```
col(k,l)   = e_(l,l+1) - e_(k,l+1)        for l <= n-2
col(k,n-1) = 2 * SUM_{j>k} e_(k,j)
```

Group rows by upper index, R_j = span{ e_(i,j) : i < j }, of dimension j. Every column with l ≤ n−2 lies in R_{l+1}, and for fixed j = l+1 the j−1 columns e_(j−1,j) − e_(k,j) span the sum-zero subspace of R_j, so their total rank is C(n−1,2). Quotient by those: R_1 is one-dimensional and untouched, each R_j collapses to its coordinate sum σ_j, and the remaining columns map to 2·(0,…,0,1,…,1) with the switch at j = k, for k = 0,…,n−2 — a triangular family, hence independent. Total rank C(n−1,2) + (n−1) = C(n,2). ■

so the rank is C(n,2), the chosen classes are independent, and dim(B_A/B_D) ≥ C(n,2) for all n. The column structure was confirmed exactly at n = 3,…,8 and the pairing matrix is invertible at every width tested up to n = 40.

5.3 The normal form

Write c_kl for the class of r_kl(2^l − 2^k) and let m(x) be the number of consecutive 1s of x starting at bit j+1.

```
[ r_ij(x) ]  =  (-1)^(x_j) * Q_ij^{m(x)}


   where   Q_ij^m  =  c_ij + SUM_{t=1..m} ( c_{i,j+t} - c_{j,j+t} )
```

An exponentially large family of basepoint-labelled squares factors through two data: the seam bit and the carry-run depth. Nothing else about x is projected. The proof is four lemmas and a telescoping recursion.

5.4 L1 — transport below the seam

The class does not depend on any bit of x below i. Let ℓ = x mod 2^i be nonzero, let k be the position of its lowest set bit, and put y = x − 2^k, so y_k = 0. The cube identity on k < i < j reads

```
r_ij(y+2^k) - r_ij(y)
```

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

```
=  [ r_kj(y+2^i) - r_kj(y) ]  -  [ r_ki(y+2^j) - r_ki(y) ]
```

Every face on the right is declared: r_ki is undeclared only when bits k..i−1 are all 1 and y_k = 0; r_kj only when bits k..j−1 are all 1, which again needs y_k = 1; and adding 2^i or 2^j changes nothing below i, so all four bases keep y_k = 0. Hence the class is unchanged, and the step strictly decreases ℓ. Iterating reaches ℓ = 0. No induction on n is used. ■

5.5 L2 at the top, and the base case

In $\mathbb{Z}/2^n$ the generator 2^{n-1} is an involution, so {y, y+2^{n-1}} is the same edge from either side and e_{n−1}(y+2^{n-1}) = −e_{n−1}(y). The two squares use the same four edges, and substituting that relation term by term gives

```
S_(i,n-1)(x) + S_(i,n-1)(x + 2^(n-1))  =  0
```

in the chain group — no quotient is involved. With L1 clearing every bit below i, bits i..n−2 pinned to 1 by the undeclared condition, and bit n−1 the only remaining freedom, the base case follows: [r_{i,n−1}(x)] = (−1)^{x_{n−1}} c_{i,n−1}. ■

5.6 The domino: the seam sign is the carry identity

Stack r_ij(x) and r_ij(x ⊕ 2^j) for x_j = 0. They share an edge, and the domino they form has the boundary of the next pair's square — up to two copies of the carry triangle:

```
S_ij(x) + S_ij(x XOR 2^j) = S_(i,j+1)(x) + T_j(x+2^i) - T_j(x)


T_j(y) = e_j(y) + e_j(y + 2^j) - e_(j+1)(y)      [ A_j^2 = A_(j+1) ]
```

an exact 1-chain identity. So the sign in the normal form is not extra data: it is the undeclared carry identity itself, seen from inside the quotient rather than counted. The residual's two columns are not independent — C(n,2) supplies the magnitudes Q_ij^m and the n−1 carry identities supply the sign on every one of them. L2 below the top reduces to a single membership, S_{i,j+1}(z) + T_j(z+2^i) − T_j(z) ∈ B_D for z_j = 0.

At j = n−1 the statement degenerates: there is no bit n, the top generator is an involution, T is not even defined, and the membership is vacuous. That is the same fact as the exact identity of §5.5, reached from the other side.

5.7 The staircase

Let z have bits i..j−1 all 1 and z_j = 0. Set u_j = z and u_k = u_{k+1} − 2^k for k = j−1,…,i — each step clears a bit that is set, so no borrow occurs — and put w = u_i = z − (2^j − 2^i), so that w + 2^j = z + 2^i. The single carry edge z → z+2^i is replaced by a staircase that clears the occupied bits one at a time and then makes the 2^j jump. Two exact identities drive it:

```
(A)   T_j(y+2^k) - T_j(y)  =  S_kj(y) + S_kj(y+2^j) - S_(k,j+1)(y)
```

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

```
                                                    for k < j <= n-2

(B)   T_j(w+2^j) - T_j(w)  =  - S_(j,j+1)(w)
```

Telescoping (A) along the staircase and applying (B) for the final jump,

```
S_(i,j+1)(z) + T_j(z+2^i) - T_j(z)
   = S_(i,j+1)(z) - S_(j,j+1)(w)
     - SUM_{k=i..j-1} [ S_kj(u_k) + S_kj(u_k + 2^j) - S_(k,j+1)(u_k) ]
```

and every term on the right is declared, by three bit facts: u_k has bit k cleared and adding 2^j never touches bit k; w_j = 0; z_j = 0. No induction on width. ■

Identities (A) and (B) were checked twice: in the arena, where 2^{n-1} is an involution, and formally in symbols (generator, offset) with offsets left as unreduced integers — no modulus and no n anywhere. 270 and 60 formal instances respectively, zero nonzero residues; the full staircase 180 formal instances with 2340 declaredness checks, zero failures; and in the arena exact with every term declared through n = 10 (3586 instances).

5.8 Assembling

The algebra telescopes. For m ≥ 1,

```
Q_ij^m  =  c_ij + Q_(i,j+1)^(m-1) - Q_(j,j+1)^(m-1)
```

because the c_{j+1,·} terms cancel between the two tails. Its geometric source is the cube identity at k = j+1, an exact 1-chain identity whose right-hand terms both have upper index j+1 > j — so the induction runs downward from j = n−1, where m is forced to 0 and the base case of §5.5 applies. With L1, L2 and the base case, the normal form is a theorem for all n. Hence B_A/B_D = span{c_ij}, so dim ≤ C(n,2); the witness of §5.2 gives ≥; and the two meet:

```
dim(B_A / B_D) = C(n,2)


b_1^disjoint(n) = C(n,2) + (n-1) = (n-1)(n+2)/2
```

5.9 The pair-module dimension

Let M_ij be the span of all (i,j)-square classes in B_A/B_D. By the normal form it is spanned by Q_ij^m for m = 0,…,n−1−j, and Q^m − Q^{m−1} = c_{i,j+m} − c_{j,j+m}. So the change of basis to { c_ij, c_{i,j+1} − c_{j,j+1}, …, c_{i,n−1} − c_{j,n−1} } is unit triangular, and that family is independent because each difference carries the generator c_{i,j+m}, which occurs in no other member, while the c_ab are a basis of B_A/B_D — there are C(n,2) of them spanning a C(n,2)-dimensional space. Hence dim M_ij = n − j for all n, and the geometric reading is exact: the pair-module's dimension is the number of possible carry-run depths above the seam.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

5.10 Verification

The normal form was checked on every undeclared square at each width to n = 8 — exhaustive per width, 1538 squares at n = 8, not a sample — at three independent primes (1000003, 999983, 7919) and exactly over $\mathbb{Q}$ with rational arithmetic for n ≤ 6. Zero deviations throughout. L1's clearing step was confirmed to have all four side faces declared at all 1044 steps at n = 8; the top identity at 4096 squares at n = 9; the domino at 960 instances at n = 7; and dim M_ij = n − j both symbolically (every pair, n = 3..12) and against measurement (n = 5,6,7).

6. What the two theorems say

6.1 A rank and a state count

dim(B_A/B_D) = C(n,2) is a rank: additive over pair directions, identical for every reader of the arena. $\log_2 F(S)$ is a state count: multiplicative over the runs of a cut, different for every reading order. They are two quotient constructions on one transition system, not two measurements of a hidden quantity, and no rescaling relates them. At n = 7 the arena has 27 unresolved relation directions while an LSB-first reader carries one bit and an even-odd reader carries $\log_2 36 \approx 5.2$ bits.

There is a second and sharper obstruction. 2L+1 is odd, so any cut containing an interior run has memory that is not a power of two: its retained state has no bit-decomposition and there is no register to count. Whenever a residual is reported as so many bits of memory, that is a statement about the cut being anchored, not about the object.

6.2 The shared sufficient statistic

static (object side)	**dynamic (reader side)**
an undeclared (i,j)-square keeps the seam bit x_j and the carry-run depth m(x)	a run keeps its first-nontransparent depth d and its carry mode
2^n basepoints collapse to (x_j, m)	4^|S| assignments collapse to a signature tuple
dim M_ij = n − j = carry depths above the seam	2L+1 = states of an interior run of length L
the residual is identical for every reader	the state count depends on the cut S

Both proofs turned on the same object from opposite ends. The exponentially large coordinate label is, in each case, the reader's own coordinates and not the object's structure.

6.3 Where the carry bit comes from

The carry recurrence c_{q+1} = g_q ∨ (p_q c_q) is intrinsic to binary addition. That one bit of retained state suffices is intrinsic to the bottom-anchored cut: the lower carry port is pinned at $c_0 = 0$, so every output inside the run is immediately emitted and nothing is owed; the only survivor is where the carry goes next. Move the anchor and the state stops being a bit.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

A system need not retain its history. It must retain exactly the cut-defined continuation structure through which future inputs can still make past distinctions matter.

7. Corrections and discrepancy log

Every item here was caught by a disagreement between two readings rather than by inspection. They are preserved because the failure modes are more informative than the results.

1. Owed-set count. Predicted L+1 owed-sets per interior run (the proper prefixes plus the empty one); measured L. The empty ledger never occurs in the interior — a partial reader of an interior run always owes at least its lowest bit — and the transparent state reuses the longest owed prefix with a different carry mode. The product 2L+1 is unchanged; the factorisation is L×2+1, not (L+1)×2−1.
2. "Middle-out" names several orders. At n = 5, reading 0 before 4 gives [1,3,5,7,2,1] and reading 4 before 0 gives [1,3,5,7,5,1]. Same peak. Not a discrepancy.
3. The orbit conjecture — that memory counts relation orbits under the reader's step — is retired, not weakened. Orbits number n; measured memory is one bit at every width.
4. A self-contradictory module claim. An earlier draft asked for rank 1 per pair-module and independence across pairs, then said in the next clause that the submodules overlap heavily. The literal reading is false: dim $M_{ij} = n - j$. The correct target was $\dim(B_A/B_D) = C(n,2)$, and the overlap is the mechanism rather than an obstacle.
5. A single prime is not a rational check. An early pass ran only at p = 1000003. Re-checked at three primes and exactly over $\mathbb{Q}$.
6. "LSB-first is the unique minimum" has an n = 2 exception. The computation's own output printed 2 of 2 orders at n = 2 while the abstract said "exactly one out of n!" flatly.
7. A basepoint convention, mistaken for a 50% failure rate. The membership test failed on exactly half the instances until based at the seam-zero point: $x \oplus 2^j = x - 2^j$ when $x_j = 1$, not $x + 2^j$.
8. A broken formal check reported 100% failure. The chain accumulator left zero-valued keys in a dictionary and a truth test on the dictionary counted them, so every identity looked violated while the arena computation said otherwise. Two readings disagreeing meant one reader was broken, and it was the new one. Fixed by testing coefficients rather than keys. This case is worth keeping: it nearly produced a live 180-of-180 refutation of a correct proof.
9. Wording. "The carry bit is not a fact about addition" was too strong. The carry recurrence is intrinsic to binary addition; what the bottom-anchored cut fixes is the dimensionality of the retained state.

Attribution. The staircase decomposition closing the last homological membership (§5.7) and the transducer argument closing the reader law (§3) were both supplied in review. They are verified here, not merely adopted.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

8. Open problems

10. The worst-order closed form. 4·3^((n−3)/2) and 20·3^((n−6)/2), DP-verified to n = 60, with an exchange sketch on run-length density but no proof. A combinatorial optimisation over run patterns, distinct from the factorisation theorem.
11. Carry-lookahead readers. The reader here receives one bit position at a time. A reader receiving prefix aggregates — generate/propagate summaries — declares different relations, so its cut algebra is a different object. This is the natural test of whether the cut law survives beyond positional readers.
12. Arenas where the two quotients agree. Characterise systems in which residual rank and $\log_2$ of the minimal state count are related by a theorem. The register rung, with $b_1 = 0$ under locality, is the trivial baseline; nothing is known between.
13. How physics selects the predicate. The locality predicate used here is canonical relative to exclusion and locality — it is Mazurkiewicz independence, and for token systems the resulting complex computes the hard-core configuration space. That a physical dynamics selects it does not follow and is not claimed. This is the only question in the programme that has not moved.

9. Methods and reproducibility

All results are produced by a single self-contained notebook, two_quotients_of_one_machine.ipynb, which accompanies this paper. It carries one library cell and imports nothing from earlier work; every number quoted above is live output from the cell beneath its claim.

Two engines are used and they share no code. The reader engine enumerates Myhill–Nerode classes by explicit completion and never mentions carries; the arena engine builds the Cayley complex and computes ranks over a prime field and never mentions readers. Homological computations run at p = 1000003, with the load-bearing normal-form check repeated at p = 999983 and p = 7919 and confirmed exactly over $\mathbb{Q}$ with rational arithmetic. Formal chain identities are verified in symbols (generator, offset) with unreduced integer offsets, independent of modulus and width.

Reproduction manifest. Python 3.11, NumPy 2.4. Homological ranks over the prime field p = 1000003; the normal-form check is repeated at p = 999983 and p = 7919 and again exactly over $\mathbb{Q}$ with rational arithmetic — all three executed in the notebook, not asserted in prose. Maximum widths: reader partition n ≤ 6 exhaustive and n = 7,8 sampled; arena b_1 to n = 8; normal form to n = 8; staircase to n = 10; the symbolic identities carry no width. The pair-module rank certificate is computed with exact modular elimination rather than floating-point rank, since the unit-triangular argument is the proof and the computation is only a check. Total runtime is about a minute.

Where a claim is verified per width rather than proved, it is labelled as such. Exactly one claim in this paper is in that category: the worst-order closed form of §4.3. Everything else stated as a theorem is proved for all n.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

References

A. Abrams, Configuration spaces and braid groups of graphs, PhD thesis, UC Berkeley, 2000. — the discretized configuration space of tokens on a graph, cited in §2.1 as external context for the locality predicate; not used in any proof here.

A. Mazurkiewicz, Concurrent program schemes and their interpretations, DAIMI report, Aarhus, 1977. — trace independence, of which the predicate here is a state-dependent analogue.

J. Myhill (1957) and A. Nerode (1958). — the right-congruence characterisation of minimal states used throughout §3.

---

v1.1 — September 2026. Consolidates the adder rung of the interface programme, superseding the v1–v5 working notebooks. v1.1 incorporates review corrections to the definitions, the Mazurkiewicz and Abrams attributions, the Claim B statement, the chain notation, and the four-case distinguishing argument.

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com